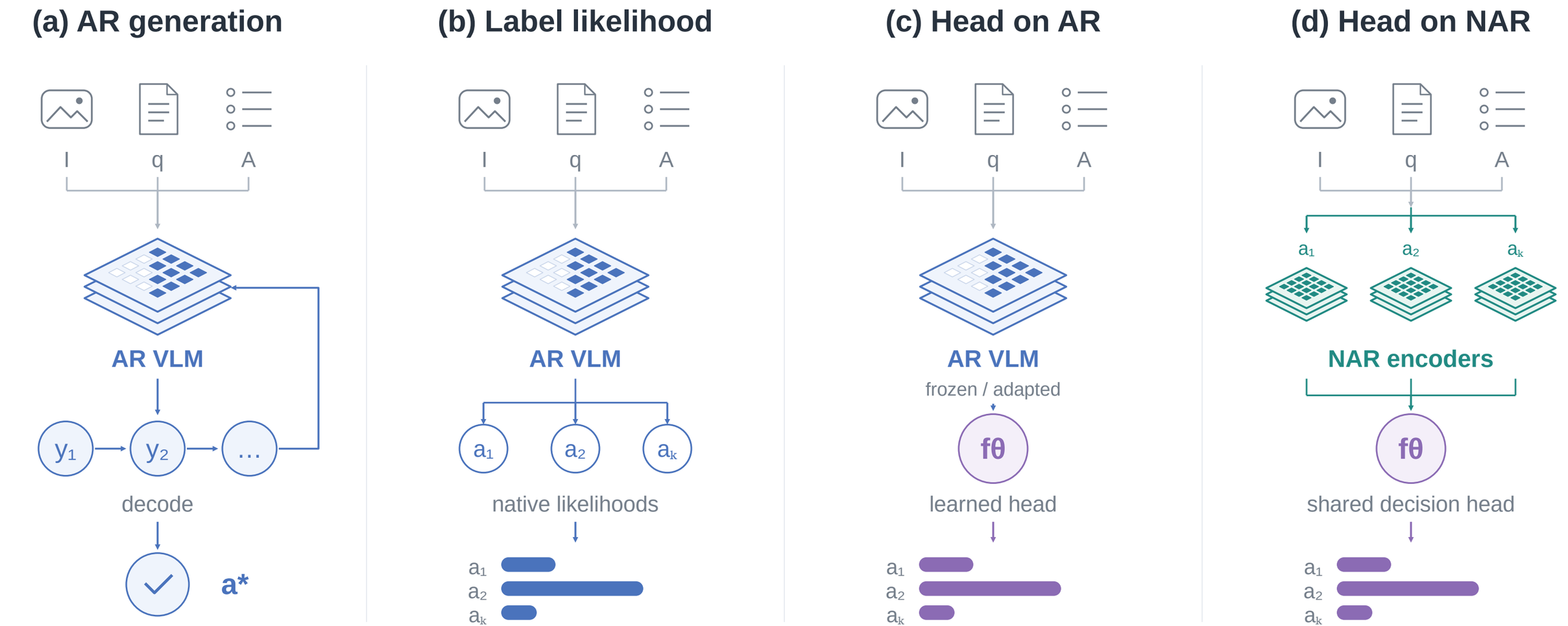

(a) AR generation
I
q
A
AR VLM
y1
y2
…
decode
a*
(b) Label likelihood
I
q
A
AR VLM
a1
a2
ak
native likelihoods
a1
a2
ak
(c) Head on AR
I
q
A
AR VLM
frozen / adapted
fθ
learned head
a1
a2
ak
(d) Head on NAR
I
q
A
a1
a2
ak
NAR encoders
fθ
shared decision head
a1
a2
ak
Same inputs in every panel: image I, question q, candidates A
AR backbone
NAR backbone
Learned head